Penlink מבית InsAlghts Academy בשיתוף Google and Richman AI Tech School

Final Technical & Executive Report

רשת המרכולים "שפע יששכר"

The Shefa Be-Paa Initiative: Orchestrating Retail Efficiency through Advanced Data Modeling

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Executive Summary

This project delivers an end-to-end, data-driven evaluation of supermarket operations, customer behavior, and checkout efficiency using geolocation and transactional data from June to November 2025. The primary objective was to transform raw operational data into actionable insights that support strategic decision-making across staffing, self-checkout deployment, customer monetization, and long-term performance monitoring.

Key takeaways for stakeholders are clear. First, customer demand and operational pressure are highly concentrated at specific temporal windows, particularly on Thursdays and Fridays, requiring differentiated staffing and automation strategies rather than uniform resource allocation. Second, the supermarket exhibits strong customer loyalty and app penetration, with nearly 90% of sales linked to app users, validating the app as a reliable analytical backbone despite its sub-benchmark geolocation capture rate. Third, revenue is structurally concentrated: a small segment of high-value customers (VIPs) generates a disproportionate share of revenue, while a large low-value segment consumes significant operational capacity. Finally, self-checkout should not be viewed as a generic cost-saving tool, but as a dynamic operational lever whose role shifts between cost optimization, capacity buffering, and peak-hour operational support.

Most importantly, the analysis addressing Question 3א stands as a central pillar of this project. By integrating customer demand, cashier capacity, and percentile-based stress testing, the study provides a defensible, hour-by-hour decision framework that translates complex operational signals into clear managerial guidance on staffing and self-checkout usage.

1. Introduction & Business Question

Modern grocery retail operates under tight margins, high labor costs, and increasingly volatile demand patterns driven by seasonality, holidays, and external shocks. In this context, relying on static schedules or intuition-based staffing decisions exposes the business to both cost leakage and customer experience degradation.

The central business question of the Penlink project is:

How can the supermarket optimally align staffing levels, self-checkout usage, and operational resources with real customer demand in order to maximize efficiency, control labor costs, and protect the customer experience?

Supporting this core question are several sub-questions:

- When is the store truly operational, based on customer and sales signals rather than assumed schedules?
- How reliable is the mobile app as a measurement tool for in-store behavior?
- Which customer segments generate the majority of revenue, and how efficiently do they use store resources?
- How does time spent in-store translate into monetization, and where do diminishing returns appear?

2. Data Methodology & Exploratory Data Analysis (EDA)

Data Sources

Two primary datasets were used:

- Geolocation: device-level location pings with timestamps, roles, areas, and spatial accuracy (accuracy_m).
- Log Sales: transactional records including sale_id, timestamp, subtotal, and customer_id.

Data Cleaning and Quality Decisions

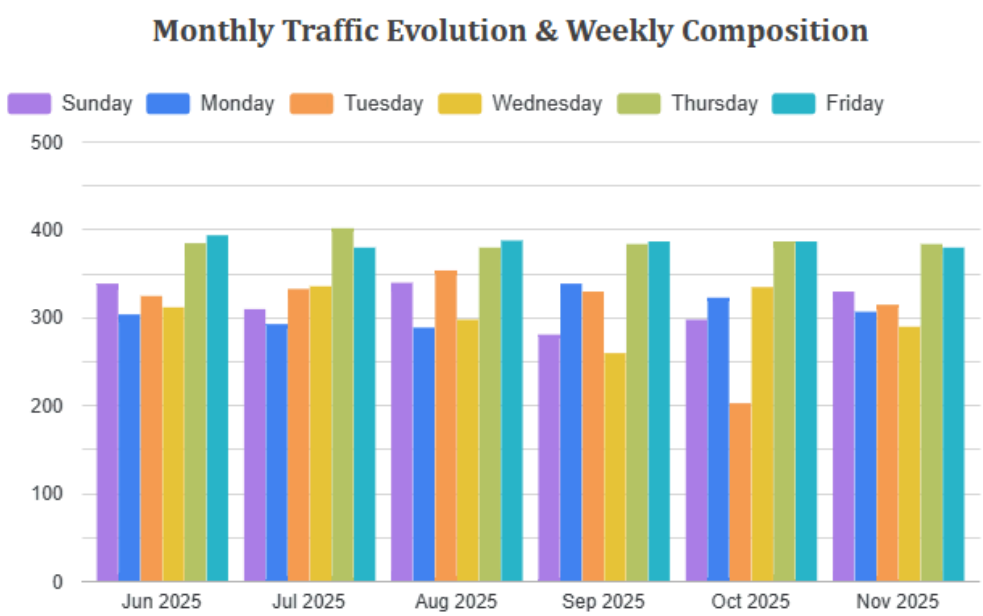
A critical methodological decision concerned the treatment of spatial accuracy outliers. Location transmissions with accuracy_m greater than 30 meters were classified as low-precision. Out of 1,308,624 total signals, approximately 31.76% fell into this category. These transmissions were excluded from subsequent behavioral analyses to prevent distortion of dwell time, zone presence, and congestion metrics.

This exclusion was justified on both statistical and operational grounds: while the proportion of low-accuracy signals is non-trivial, their inclusion would systematically bias in-store analyses by introducing noise from parking areas and transitional zones. Importantly, role-level analysis revealed that higher outlier rates (e.g., security staff in parking areas) were consistent with operational context rather than application failure.

Handling Temporal Outliers

Employee arrivals slightly before opening hours and departures after closing were intentionally retained. These observations reflect real operational processes (opening, closing, cash reconciliation, security) and do not represent data errors. Notably, no temporal outliers were observed on Wednesdays, indicating unusually stable operations on that day.

Exploratory Traffic Patterns



This chart provides a detailed view of store traffic trends from June to November 2025.

EDA revealed a stable baseline of 300-400 customers per day, with pronounced and consistent peaks on Thursdays and Fridays across all months. A notable anomaly was observed in October 2025, where Tuesday traffic dipped sharply, highlighting the importance of monitoring weekday-specific volatility rather than relying on weekly averages alone.

3. Deep Dive Analysis – Operational and Behavioral Insights (Question 3א)

Customer Behavior and App Engagement

Repeat customers dominate store engagement, accounting for over 60% of geolocation activations, while one-time customers represent less than 2%. This confirms a highly loyal customer base. Combined with an app penetration rate of 89.6% of all sales, the mobile app emerges as a strategically reliable instrument for behavioral and transactional analysis.

Validation of App Performance

Although the app captures only ~32% of expected geolocation transmissions (below the designed 40% benchmark), internal validation shows that the derived dwell_minutes metric captures approximately 88% of observed in-store presence. This indicates partial capture but internal consistency, sufficient for aggregated behavioral analysis.

Parking activity was shown to inflate dwell time by approximately 4.9 minutes (10.5% of total dwell), reinforcing the decision to exclude parking signals for in-store efficiency metrics.

Store Operating Hours and Anomaly Detection

A customer-driven definition of store operating hours was established by cross-validating sales and geolocation signals. All days without sales coincided with zero customer presence and were confirmed as Saturdays or official Israeli holidays. Two days of extreme activity aligned with documented external security events, demonstrating the importance of contextual validation.

Jun 1, 2025 - Nov 30, 2025

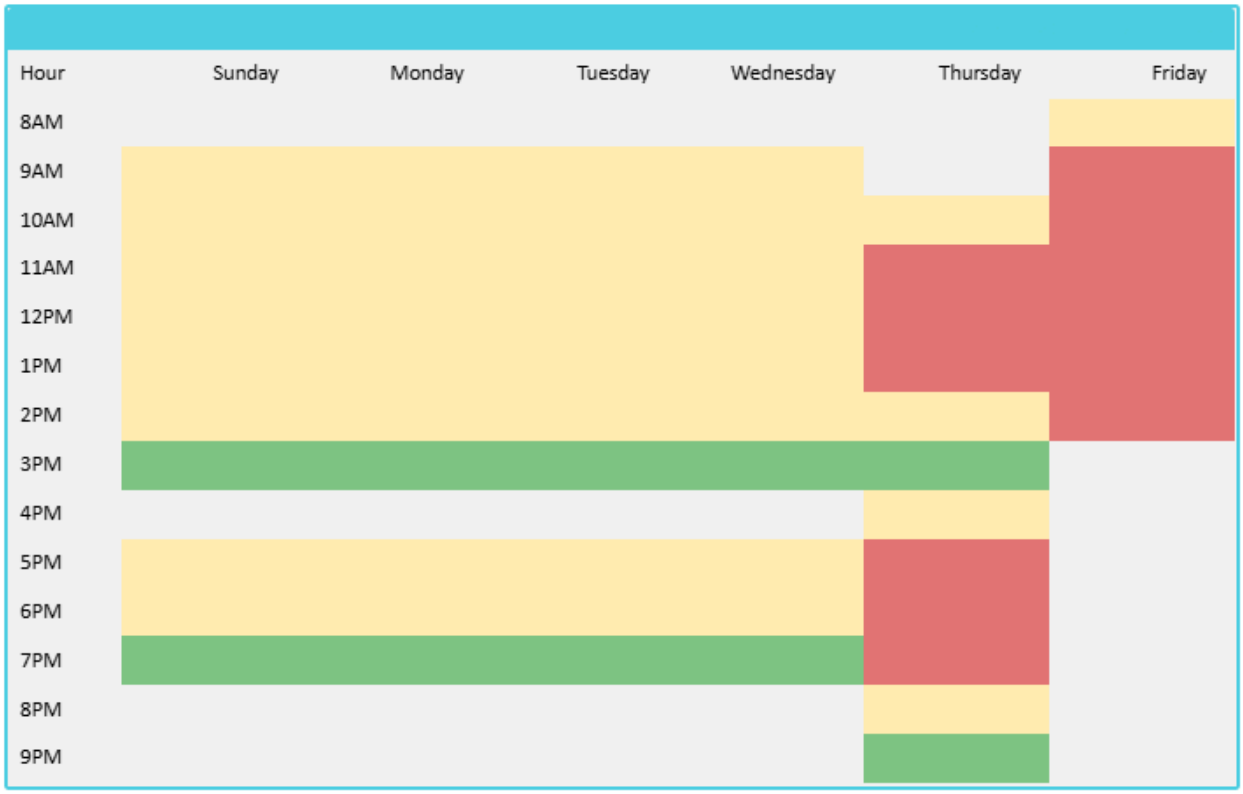
Customer Traffic Density & Behavioral Profiling

HOUR	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday
8AM	-	-	-	-	-	604.31
9AM	287.96	280.48	282.35	267.32	-	1,419.5
10AM	259.33	245.56	273.17	231.12	779.92	984.19
11AM	289.22	284	284.96	272.04	1,053.64	736.73
12PM	273.89	248.28	252.83	229.44	985.68	1,401.65
1PM	295.3	266.68	318.96	284.08	1,048.2	1,613.54
2PM	250.78	224.28	260.91	253.76	195.52	1,004.62
3PM	10.44	15.25	13.8	11.33	2.38	-
4PM	-	-	-	-	791.96	-
5PM	274.52	276.72	297.74	285.88	1,043.16	-
6PM	246.63	246.92	269.43	275.12	972.96	-
7PM	13.16	11.16	11.22	12.4	1,063.92	-
8PM	-	-	-	-	183.32	-
9PM	-	-	-	-	2	-

Peak vs. Dead Hour Identification (Core of Question 3א)

A percentile-based composite activity index was constructed by combining customer presence (60% weight) and transaction volume (40% weight). This approach avoids arbitrary thresholds and reflects real retail pressure more accurately than single-metric models.

Weekly Peak & Off-Peak Analysis: Footfall vs. Transaction Trends



Interpretation of Results

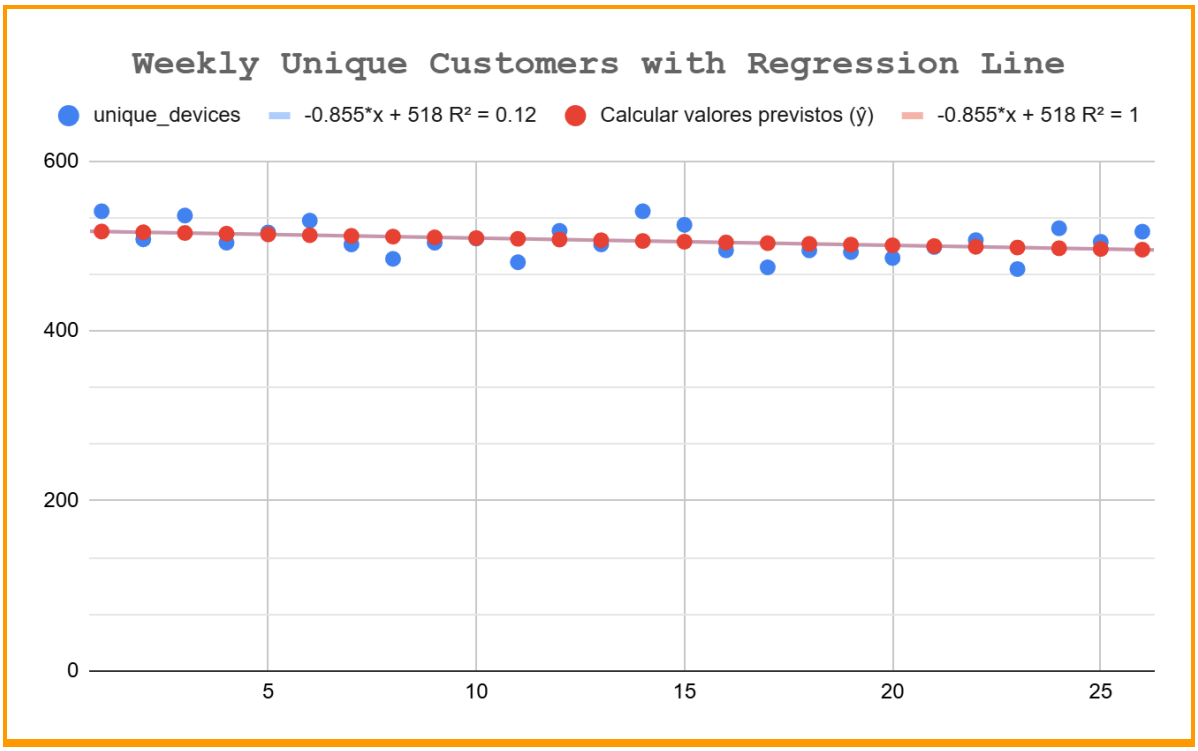
- Peak hour (≥ P75): High customer density *and* strong checkout activity → maximum operational load.
- Dead time (≤ P25): Low traffic and low transactions → minimal operational pressure.
- Regular hour: Balanced, manageable flow.

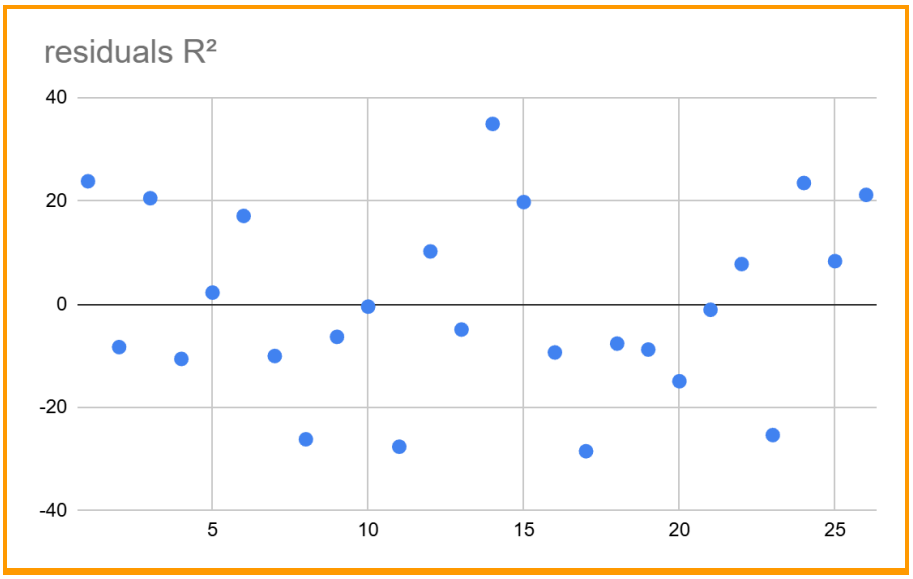
Results show that peak operational windows are heavily concentrated on Thursday and Friday afternoons, while consistent dead hours appear around mid-afternoon and early evening on other weekdays. This finding underpins all subsequent staffing and self-checkout recommendations.

4. Regression Models

Regression 1: Weekly Unique Customers Over Time

A linear regression of weekly unique customers against time revealed a slight negative trend ($\beta_1 = -0.85$), with an R^2 of 0.12 and a two-tailed p-value of 0.083. While not statistically significant at the 5% level, the result serves as an early diagnostic signal rather than evidence of structural decline. Time alone explains little variance, reinforcing the need for multivariate analysis.





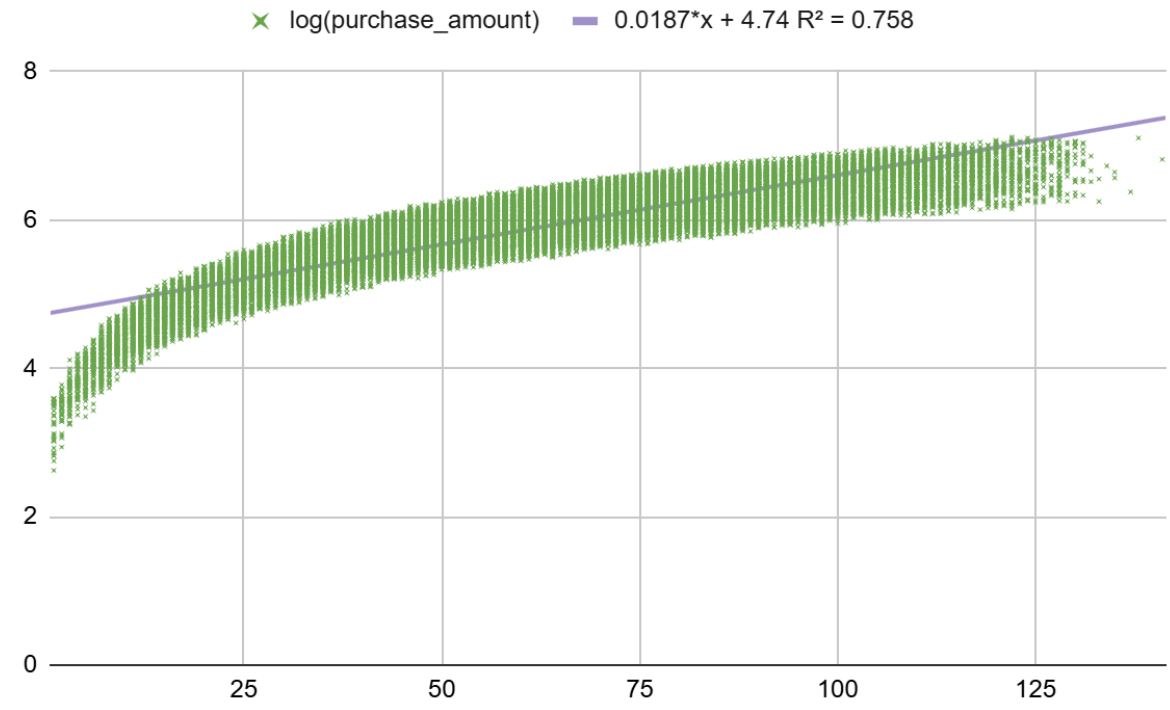
Regression 2: Purchase Amount vs. Dwell Time

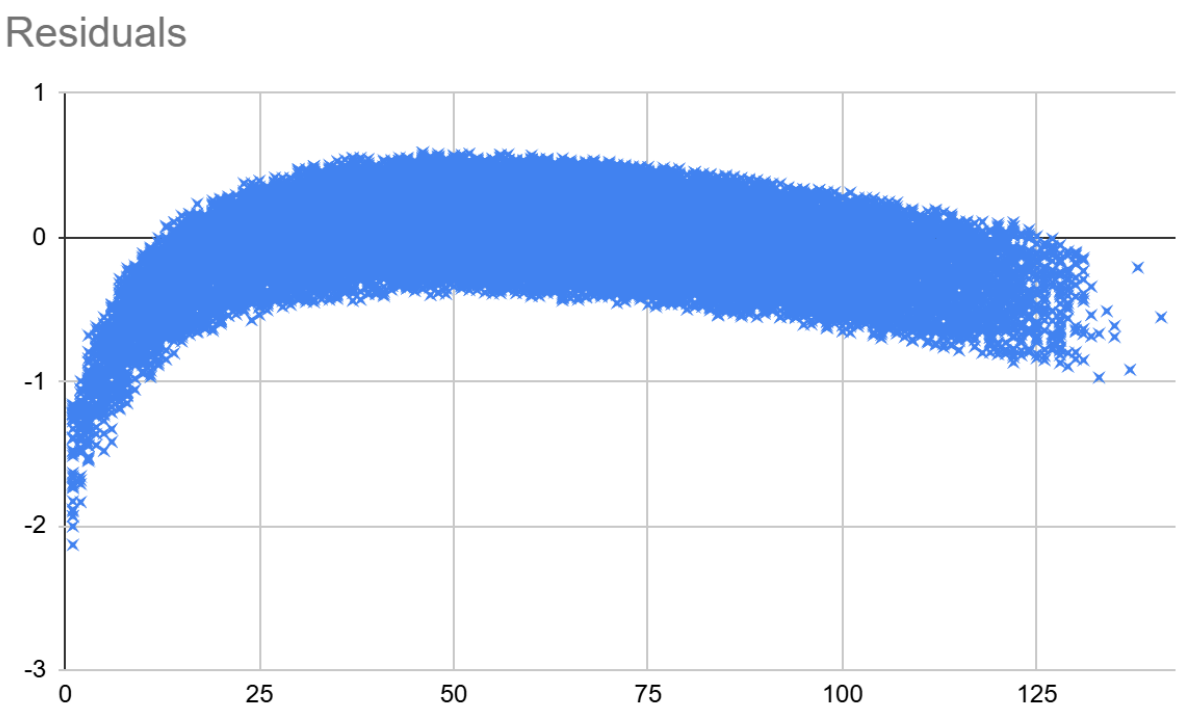
A baseline level-level regression showed a strong positive relationship ($\beta_1 = 6.46$, $R^2 = 0.759$, $p < 0.001$), but residual analysis revealed heteroscedasticity. A log-linear model corrected variance instability, yielding an interpretable elasticity of 1.87% increase in purchase value per additional minute in-store.

However, residual curvature indicated non-linearity, leading to a business-driven time segmentation (Quick Trip, Regular Shopping, Long Stay). Segment-level regressions demonstrated diminishing marginal returns for long visits, highlighting that more time does not always equate to proportional revenue.

This Scatter Analysis maps the 'Monetization Velocity' of the store. By cross-referencing Dwell Time with Ticket Value, the analysis identifies that VIPs provide the highest ROI per minute

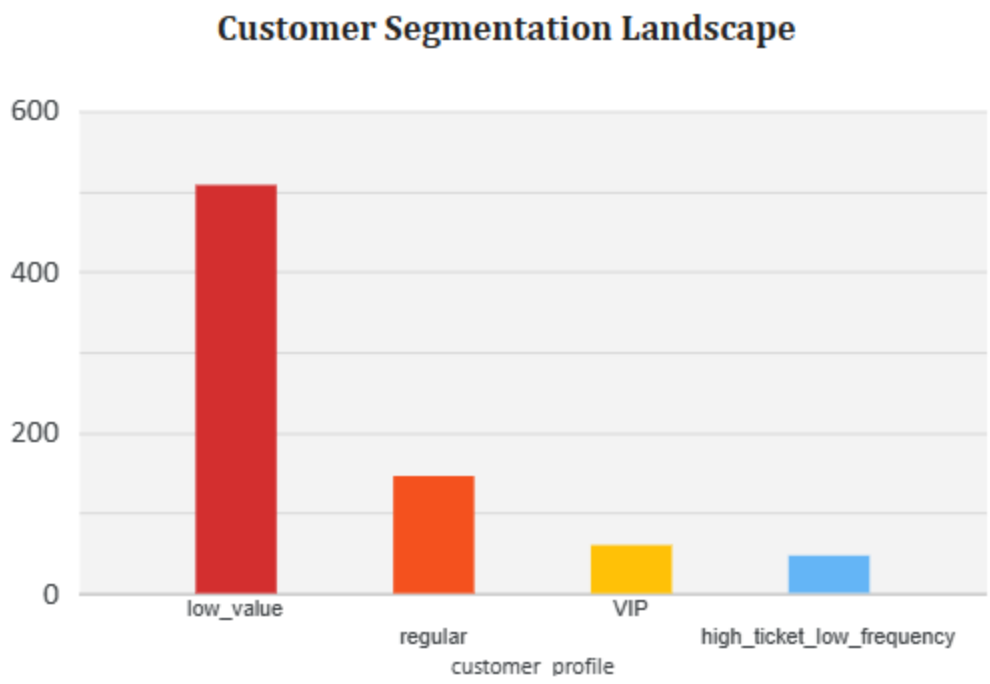
spent on the floor. The quadrants highlight segments that stay too long with low returns, signaling a need for self-service optimization.





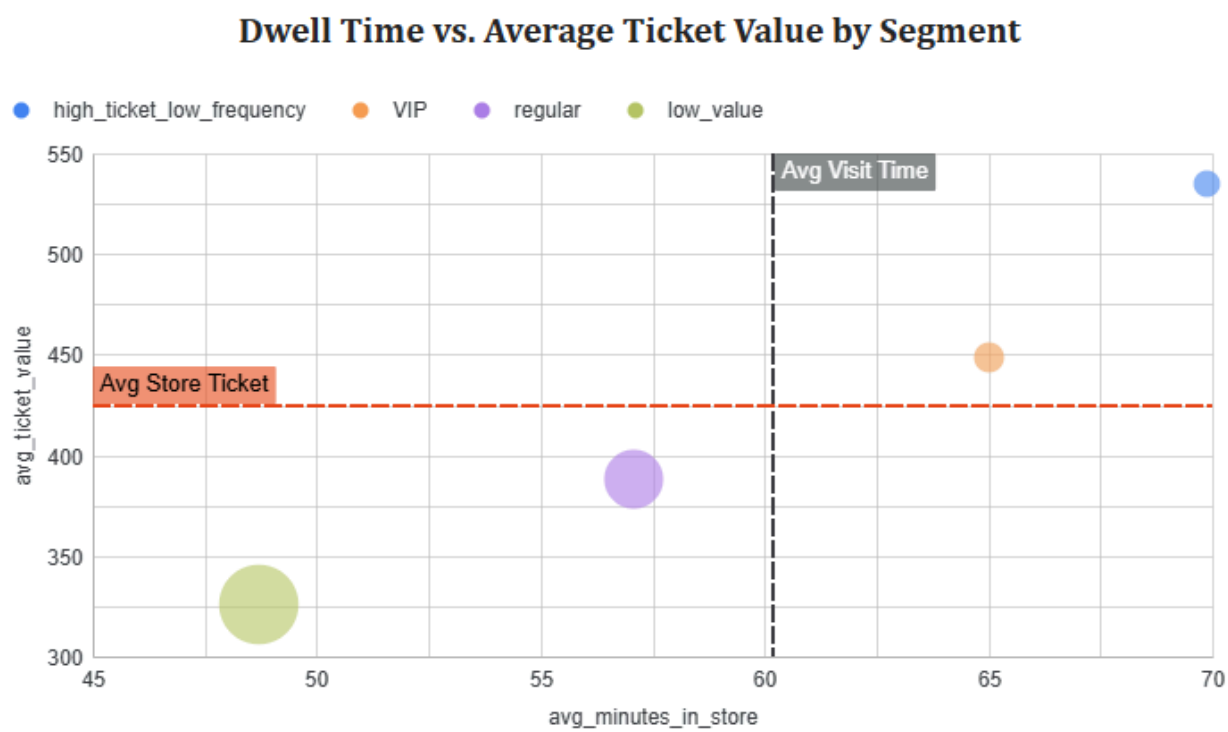
5. Customer Financial Segmentation and Monetization Efficiency

Using dynamic percentile thresholds (P25/P75), customers were segmented by average ticket and visit frequency into five strategic profiles: VIP, High-Ticket/Low-Frequency, Low-Ticket/High-Frequency, Regular, and Low Value.



VIPs represent only 7.88% of customers but generate 17.94% of revenue, confirming a classic Pareto-like concentration. Low-Ticket/High-Frequency customers contribute materially to revenue but exert disproportionate operational pressure, making them ideal targets for self-checkout optimization.

Efficiency analysis using Revenue per Minute (RPM) showed that VIPs deliver the highest monetization velocity, while certain segments consume time and space without proportional returns.

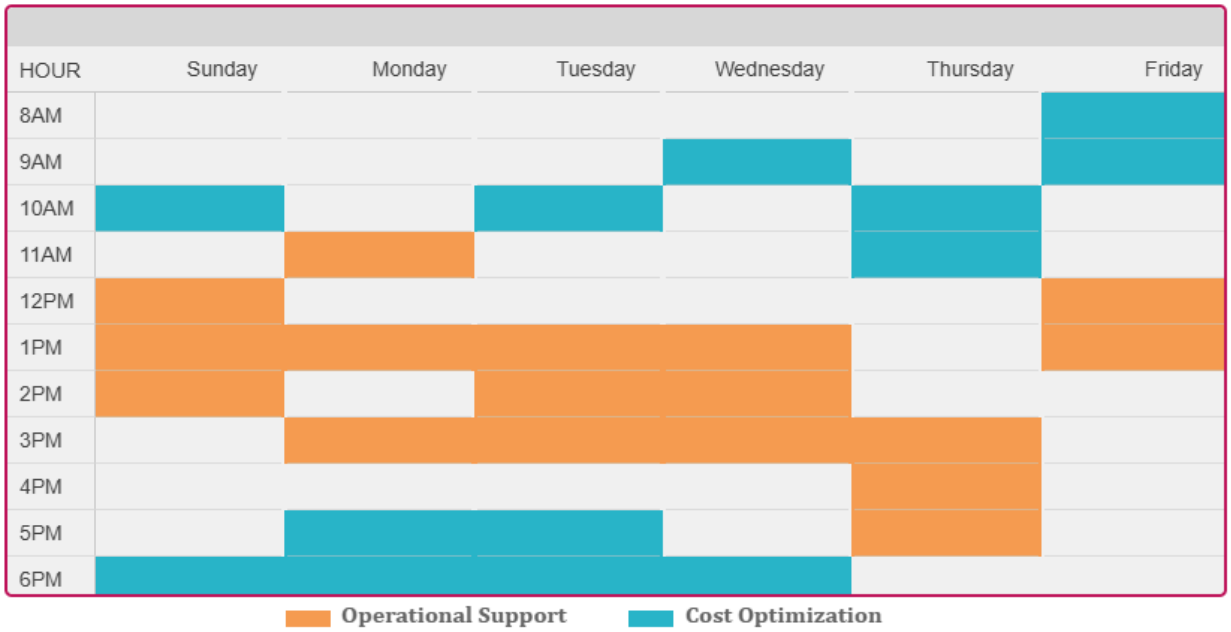


6. Staffing, Demand, and Self-Checkout Optimization (Question 3א Highlight)

The integrated staffing × demand × self-checkout KPI combines p75 demand stress testing, observed cashier productivity, and percentile-based pressure classification. Each hour is translated into a clear self-checkout role:

- Operational Support during peak and understaffed conditions.
- Cost Optimization during dead times.
- Flex Capacity Buffer during regular hours.

Self-Checkout Operational Decision Matrix



This framework converts complex analytics into manager-ready guidance, enabling proactive staffing and automation decisions.

Comparative Analysis: Footfall Dynamics vs. Checkout Pressure

1. Weekday Mid-Day Regularity (Mon–Wed)

The data identifies a consistent "Regular Hour" pattern between 12:00 PM and 2:00 PM across most weekdays. While footfall (total customers in-store) remains within manageable levels, the Pressure Index at checkout reaches a local peak.

- Insight: This indicates a concentrated "Lunch Hour" mission where customers have lower dwell times but high checkout concurrency.
- Operational Stance: Staffing levels are generally sufficient ("Well-Staffed"), but this window serves as the primary daily test for checkout throughput.

2. The Thursday Pre-Weekend Surge (3:00 PM – 7:00 PM)

A critical shift occurs on Thursdays, where the p75 Demand Model signals sustained "Peak Hour" classification from 3:00 PM to 7:00 PM. Unlike the quick mid-day peaks, this is a multi-hour high-pressure event.

- Footfall vs. Transaction: We see a "piling effect" where footfall grows steadily throughout the afternoon, leading to significant pressure on manual cashiers.

- Staffing Gap: This period shows a frequent "Understaffed" status. It is the most vital window for Operational Support, where Self-Checkout (SCO) must be utilized at maximum capacity to prevent long queues that damage customer satisfaction.

3. Friday: High-Intensity Resilience (All Day)

Friday stands out as the most volatile day for the Yavne branch. Footfall remains at Peak/Off-Peak high levels throughout the entire operating day, driven by pre-Shabbat shopping.

- Critical Pressure Window (12:00 PM – 1:00 PM): Despite high traffic all day, the Pressure Index hits its absolute maximum between 12:00 PM and 1:00 PM.
- The "Friday Paradox": Transactions are high, but the staff_gap is most pronounced here. Because footfall is sustained, any delay at checkout causes an exponential increase in store density.
- Strategic Recommendation: Friday requires an "All-Hands-On-Deck" approach. The analysis justifies the use of SCO not just for cost-saving, but as a critical Resilience Buffer to maintain store flow during the final shopping hours before closure.

Staffing analysis with peak / dead hour classification

HOUR	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday
8AM	-	-	-	-	-	
9AM					-	
10AM						
11AM						
12PM						
1PM						
2PM						
3PM						-
4PM	-	-	-	-		-
5PM						-
6PM						-
7PM						-
8PM	-	-	-	-		-

7. Business Insights & Recommendations

1. Adopt Dynamic Staffing Models: Shift from fixed schedules to demand-driven staffing, with reinforced coverage on Thursdays and Fridays and leaner operations during structurally dead hours.
2. Protect and Monetize VIP Customers: Prioritize queue reduction and premium experience for high-value customers, as the financial impact of losing a single VIP is disproportionately high.

3. Deploy Self-Checkout Strategically: Use self-checkout as a resilience buffer during peaks and a cost-reduction lever during low-demand periods, rather than as a one-size-fits-all solution.

8. Conclusion

The Penlink project demonstrates how rigorous data engineering, statistically robust methods, and business-aware interpretation can transform raw operational data into strategic intelligence. By grounding every decision in observed behavior and contextual validation, the analysis provides a defensible foundation for future operational planning.

Most critically, the solution to Question 3א establishes a scalable decision framework that aligns staffing, self-checkout investment, and customer demand in real time. This approach equips management with the analytical clarity required to operate efficiently in an increasingly complex retail environment, ensuring that future decisions are proactive, data-driven, and economically justified.